



TRANSLATION HANKOOK

PORSCHE Inspection Catalog Chassis: Tires

Testing Procedure F.360.305

Brakes Dry

Test Carrier:	Vehicle
Testing Facility:	Test Track
Status (Porsche):	18.01.2017
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(see target data sheet / "Zieldatenblatt")	

1 Purpose

This test method defines the procedure used to determine the deceleration forces of tires on a dry track. The evaluation criteria are:

- window braking from 100km/h to 1km/h (start from 105km/h)
- window braking of 70-20km/h (start from 105km/h)

The window braking distance from 100km/h and the window braking evaluation from 70-20km/h is a relevant test for tire approval and should also be used by the tire manufacturer during iteration loops to document development progress.

2 Field of Application

The test procedure is valid for:

- Approval Tests
- Series Monitoring
- Press tests of Summer Tires

3 Definition of Terms

Technical Braking Distance:

The technical braking distance describes the real measured braking distance including all influences such as threshold time (response time, pressure build-up), wheel load shift, excluding the reaction time of the driver. The aim is to determine preferably the same braking distance as measured in vehicle-related tests. A brake trigger must be used for the measurement of technical braking distances.

Window Braking:

In window braking, the goal is, if possible, to determine only the tire without the influencing factors described in technical braking distance. For the window braking distance, the brakes are applied as accurately as possible at 105km/h according to GPS. To determine the braking distance, the speed is integrated once from 100km/h to 1km/h and 70km/h to 20km/h to the braking distance.

Cold Braking:

In this procedure, "cold braking" refers to braking in which the brake is only at a "normal" operation temperature ($T \leq 150^{\circ}\text{C}$). For this reason, there is a longer cooling phase after each braking operation in the test to give the brake and the tires sufficient time to regenerate to the "normal" operating temperature.

4 Responsibility and Scope of Testing

Approval Test:

The test for approval is carried out and documented by Porsche. The results are made available to the tire manufacturer if required.

Retesting:

Retesting of approval tires can be arranged by the Porsche specialist department if required.

5 Description of the Testing Procedure

5.1 Measuring Equipment

5.1.1 Measured Variables / Assessment Variables

Measuring System:

Race Technology Speedbox & PSM measurement technology

If available: measure with PSM measurement technology

- Vehicle Speed in km/h
- Longitudinal Acceleration in m/s^2
- Lateral Acceleration in m/s^2
- Brake Distance in m

- Position (GPS)
- Pedal Input (Trigger) - optional

Recorded values at the beginning of each test series:

- Tire Pressure
- Temperature Asphalt (infrared temperature sensor in front of the right front wheel)
- Ambient Air Temperature in the shade
- Wind Direction and Strength if available (from weather station), maximum values only

5.1.2 Test Track

Sufficient long, dry braking track on homogeneous asphalt surface with high friction, preferably:

- High-Speed oval in Rome or Papenburg
- Braking track at Contidrom

Other approved test tracks are:

- Brake track at Idiada
- Brake tracks at Nardo (dynamic driving area)
- Brake track at Hockenheim (dragster track)
- Brake track at Boxberg
- Brake tracks at Fontange/Ladoux
- Lahr (air field)

5.1.3 Data Processing

Evaluation Software: Race Technology V8.5 (or newer)

Presentation of Results: intern PAG Excel evaluation sheet

5.2 Test Conditions

- The tire age should be between 4 to 28 weeks
- Asphalt temperature for summer tires: 14° C to 55° C
Asphalt temperature for all-season tires: 8° C to 55° C
Maximum asphalt temperature difference between two reference tires: 10° C
- Air temperature for summer tires: 14° C to 34° C
Air temperature for all-season tires: 8° C to 34° C
- Exception temperatures for BIC brakes:
Asphalt temperature for summer tires: 14° C to 45° C
Maximum asphalt temperature difference between two reference tires: 10° C
- Exception temperatures for 2U approved press tires for window braking 100-1 km/h:
 - Extended temperature window of asphalt temperature for the selection of press tires for the transition period between 0° C and 7° C
 - Extended temperature window of air temperature for the selection of press tires for the transition period between 0° C and 10° C
- Make sure that the GPS signal is not disturbed by shadowing (e.g. bridges, signs)

- The tires must not be exposed to direct sunlight before the test and they should be at approximately ambient air temperature
- The brake track and the vehicle must be preconditioned with braking on a summer rolling tire until the braking distance remain constant (at least five braking operations). If the braking distances scatter too much, the braking point must be shifted.
- The new braking system shall be braked in accordance with the latest braking program for the installed braking system of the test vehicle from the EFB specialist department.
- The settings of the measuring equipment must be checked before starting the test.

If any of the test conditions cannot be met, consult with BTV.

5.3 Test Carrier / Test Set-Up

- Set tire pressure, if available to comfort pressure "160km/h" or "270km/h" of the vehicle, else consult BTV
- Chassis adjustment: set PASM to Normal-Mode
- Air suspension: normal level
- The load condition is to be defined in advance by the client and set by the tester.
Standard: driver + measurement equipment
- ET, rim width, diameter and contour at the rim seat of the test rim must correspond th the respective original Porsche wheel (see target data sheet)
- Static vehicle data (e.g. toe-in, camber, heights) must be checked for correctness in advance and corrected if necessary
- In preparation, new or appropriately good preconditioned discs and brake pads must always be installed
- For reasons of comparability, new tires must be used. For the BIC brake series, the tires must be preconditioned according to 5.4

5.4 Preconditioning, no later than 2E 3 (BIC)

Reference and test tires are braked 25 times on the day before the test program, analogous to the procedure described in 5.5. The reference tires are to be conditioned one after the other and recorded with the measuring equipment. The drift of the reference tires must be determined. Based on the braking distance measurement data, pairings of the reference tires for the test programs are to be formed. If reliable information on the braking behavior of the reference tire is available, the conditioning of the reference tires can also be shortened to the necessary number.

5.5 Test Procedure

- Special care must be taken when setting the air pressure. If the air pressure in the tire has to be changed, it must be checked again after at least one minute, as thermodynamic effects must be compensated.
- The test speeds are based on the value measured by GPS and not on the speedometer value.

- Between each braking maneuver, a sufficient cool-down drive must be performed at a driving speed in the oval of at least 80 to a maximum of 120km/h and a lateral acceleration no greater than 3m/s^2 . The driving distance is at least 1km and not more than 6km long.
- The brake should always be applied with the same force and the same motion sequence.
- The same number of braking operations should be performed on all tire sets in a test program. The time frame should be as equal long as possible for each set of tires. The time sequence between the individual braking operations should be constant.
- All tires (reference and development tires) in a program must have the same initial condition. Either all must be new or all must be preconditioned according to the same program.
- Recommendation: Depending on the temperature range and track opening times, a test program should not consist of more than three to four development tires, two reference tires and one conditioning run.
- A reference tire set is run at the beginning and a second reference tire set with the same DOT if possible at the end of the measurement program. In the even of a major interruption (longer than one hour, lunch break), the measuring program must be completed with a reference tire set before the interruption. When resuming or in case of an unplanned interruption, a reference tire set must be measured again. An interruption should always be avoided.
- The test series must be started with a full tank. Do not refuel during a test series, or if necessary after each set.
- The tests described (see Test 1 and 2 below) can be combined as desired by the client. The sequence specified by the client must be strictly adhered to.
- **Porsche internal only:** All measurements must be recorded on a memory card and stored on the group drive. For this purpose, the files must be clearly named (record number, test sequence).

Test 1: ABS cold breaking from 100km/h

- Use new tires or tires preconditioned according to 5.4
- Starting speed: **105km/h**, if possible with cruise control set according to GPS speed
- A total of 13 full braking tests are performed with each set of tires to be tested:
 - Perform three conditioning braking operations at the defined braking point
 - Ten full braking operations at the defined braking point from the respective initial speed with ABS. With manual transmission, start braking and disengage approximately one second later; with PDK/automatic transmission, continue in gear until the vehicle comes to a standstill
- At least one second before braking, the steering wheel angle must be set to 0°

5.6 Exclusion Criteria

- If the reference tires in the initial and final measurement of a measurement program are more than 2% apart or the coefficient of variation in the Excel evaluation is red, the measurement data must be checked for plausibility and suitable substitute measures must be defined together with the BTV.
- In case of test tires, the cause must be clarified.

6 Documentation

6.1 Data Analysis

The data is evaluated using the PAG internal Excel evaluation sheet. Detailed instructions can be found in the Excel evaluation attachment.

Cold break measurements with ABS

- Presentation of Results: Analysis of 100-1km/h and 70-20km/h

Window braking results (70km/h to 20km/h) are extrapolated from 100km/h to 0km/h and averaged. The braking distance of the reference tire is set to 100%.

Based on the braking distances of the reference tires, a linear trend adjustment is performed over the measurement series. Clearly incorrect measurements must not be included in the evaluation.

6.2 Presentation of the results

In the form of the PAG Excel evaluation sheet.

6.3 Requirements

The tire must achieve the values specified in the target data sheet.

7 Applicable Documents

Appendix 1: Excel evaluation

8 Distributor

EFF

Tire Development Partner

PES